

Fixed Arithmetic Tests and the Original Heat Trace

Split-Zero research programme

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This addition carries every original heat-collision jet into fixed arithmetic Cauchy tests. It evaluates the complete next trace coefficient, the exact change of basis and local determinant, and the compensating matrix retained in the full arithmetic formula. All original units, coordinates, support labels and pole locations are kept. The complete preceding proofs HC, HH, HD, HF and HL are included as the two retained LaTeX sources in this package. HC and HH are already published in the pinned collision-holonomy edition. The new proofs are given in full below.

1 Fixed arithmetic tests on every original heat flag

The preceding flag calculation uses specified coefficient lifts in the original algebra. Here the tests are fixed functions of the original arithmetic variable. Their higher Taylor coefficients are retained. This constructs the exact map from those tests into the entire original collision algebra, and evaluates its determinant and the next trace coefficient. The heat convention is Brad Rodgers and Terence Tao, *The de Bruijn–Newman constant is non-negative*, author equations `hoz`, `htdef`, and `sas`. The full reflected arithmetic trace and support coordinates are those of SZW19–38, which proves the explicit formula with both endpoints using Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Appendix II, Theorem 6. The complete finite calculations below do not require moving an infinite zero sum through a derivative.

HR1. Original objects and fixed tests. Fix an actual real zero x_0 of the original H_0 , with its actual order m . Put $\rho = 1/2 + ix_0/2$, $s = \rho + i\xi/2$, $t = u^2$. Retain $H_t(x_0 + \xi) = U(t, \xi)W(t, \xi)$, the original unit U , and every root $\theta_\nu(u)$ of W . The complete proofs HC1–14 give

$$\theta_\nu = r_\nu u + 2cu^2 + br_\nu u^3 + O(u^4), \quad c = a_{m+1}/a_m, \quad d = a_{m+2}/a_m, \quad b = 2d - c^2, \quad (\text{HR1})$$

where r_ν are all roots of P_m in HF2. Write $p_n = \sum_\nu r_\nu^n$, so $p_0 = m$, $p_{2j+1} = 0$, and all even moments are evaluated by HF7. For fixed holomorphic functions F, G near ρ and their reflected neighbourhoods, put

$$F^\#(s) = \overline{F(1 - \bar{s})}, \quad f_n = F^{(n)}(\rho)/n!, \quad g_n = G^{(n)}(\rho)/n!, \quad Z_{\rho,t}(F^\#G) = \sum_{\nu=1}^m (F^\#G)(\rho + i\theta_\nu/2). \quad (\text{HR2})$$

Reflection fixes ρ , and sends the local variable ξ to $\bar{\xi}$. Therefore the pullbacks of F, G have coefficients $\alpha_n = (i/2)^n f_n$ and $\beta_n = (i/2)^n g_n$, and the reflected product has coefficient $\sum_{a+b=n} \bar{\alpha}_a \beta_b$ at ξ^n . The sum HR2 is analytic in t : it is invariant under $u \mapsto -u$, and each summand is holomorphic in u . It also equals the regular multiplication trace in the original algebra, by evaluation for nonzero t and analytic continuation to zero. This proves the test-to-algebra map and its reflected trace compatibility.

HR2. The next coefficient for all fixed vanishing tests. Suppose the actual Taylor coefficients of F, G vanish in degrees less than j , where $0 \leq j < m$. Then

$$Z_{\rho,t}(F^\#G) = \frac{t^j}{4^j} [p_{2j}\bar{f}_jg_j + t\mathcal{C}_j(F, G) + O(t^2)], \quad (\text{HR3})$$

where the full coefficient, without dropping any test jets, is

$$\begin{aligned} \mathcal{C}_j(F, G) = & (2jb p_{2j} + 4j(2j-1)c^2 p_{2j-2}) \bar{f}_j g_j \\ & + i(2j+1)cp_{2j}(\bar{f}_{j+1}g_{j+1} - \bar{f}_{j+1}g_j) \\ & + \frac{p_{2j+2}}{4}(\bar{f}_{j+1}g_{j+1} - \bar{f}_jg_{j+2} - \bar{f}_{j+2}g_j). \end{aligned} \quad (\text{HR4})$$

For $j = 0$ the first line is zero; no negative-index moment is used. To prove this, besides HF8 use the direct expansion of HR1:

$$s_{2j+1}(t) = 2(2j+1)cp_{2j}t^{j+1} + O(t^{j+2}), \quad s_{2j+2}(t) = p_{2j+2}t^{j+1} + O(t^{j+2}). \quad (\text{HR5})$$

All further powers contribute $O(t^{j+2})$, including their convergent analytic tail: on the finite branches $|\theta_\nu| \leq C|u|$, the tail is $O(u^{2j+3})$, and its branch sum is even in u . The three required product coefficients are exactly

$$\frac{\bar{f}_jg_j}{4^j}, \quad \frac{i(\bar{f}_jg_{j+1} - \bar{f}_{j+1}g_j)}{2 \cdot 4^j}, \quad \frac{\bar{f}_{j+1}g_{j+1} - \bar{f}_jg_{j+2} - \bar{f}_{j+2}g_j}{4^{j+1}}. \quad (\text{HR6})$$

Substitution proves HR3–4. When $j = 0$ these formulas give the complete first derivative, including the derivative of the analytic unit. When $j = 1$ the leading response is $m(m-1)\bar{F}'(\rho)G'(\rho)/2$, agreeing with HF6 under the exact coordinate $s - \rho = i\xi/2$. At every j the first nonzero coefficient is $p_{2j}\bar{f}_jg_j/4^j$; it is not a value assigned to the degenerate endpoint trace.

HR3. A complete original Cauchy-test frame. Take any m distinct poles $z_1, \dots, z_m$ with $\Re z_l > 1$, and set $F_l(s) = 1/(s - z_l)$. Let $C_\xi(t)$ be the matrix of multiplication by the original ξ in the ordered basis $1, \xi, \dots, \xi^{m-1}$ of $A = \mathbb{C}\{t\}[\xi]/W$. The exact coefficient column of the fixed test is

$$R_l(t) = \left((\rho - z_l)I + \frac{i}{2}C_\xi(t) \right)^{-1} e_0, \quad R(t) = (R_1(t), \dots, R_m(t)). \quad (\text{HR7})$$

Here e_0 is the first coordinate vector, not supported zero. At $t = 0$, C_ξ is nilpotent and $\rho - z_l \neq 0$, so all inverses and their analytic continuations exist. Formula HR7 follows by multiplying by the denominator in A ; it retains all coefficients of W at every time. In particular,

$$R_{nl}(0) = \frac{(-i/2)^n}{(\rho - z_l)^{n+1}}, \quad \det R(0) = \frac{(-i/2)^M \prod_{a < b} (z_b - z_a)}{\prod_l (\rho - z_l)^m}, \quad M = m(m-1)/2. \quad (\text{HR8})$$

Indeed the first formula is the finite geometric inverse; the second is the Vandermonde determinant in $1/(\rho - z_l)$. Its nonzero value proves that the fixed tests give an analytic basis of the entire original coefficient algebra as a module. This is a module isomorphism, with the algebra multiplication transported by HR7 when needed.

There is also an explicit fixed rational test realizing each arithmetic endpoint jet. Put $D(s) = \prod_l (s - z_l)$, and let $[D(\rho + v)]_{< m-j}$ mean its Taylor polynomial of degree at most $m - j - 1$. Then

$$B_j(s) = \frac{(s - \rho)^j [D(\rho + v)]_{< m-j}|_{v=s-\rho}}{D(s)} = (s - \rho)^j + O((s - \rho)^m), \quad 0 \leq j < m. \quad (\text{HR9})$$

The numerator has degree at most $m - 1$, so $B_j = \sum_l N_j(z_l)F_l/D'(z_l)$, by subtraction of its simple-pole residues and decay at infinity. The Taylor remainder in the numerator is $O(v^m)$ and $D(\rho) \neq 0$, proving HR9. Thus every endpoint jet is reached by actual fixed Cauchy tests. Their coefficients at degrees m and above remain those of HR9 and must be included in HR4; they are not set to zero.

HR4. Exact Gram determinant and its arithmetic correction. Write $K(t) = (s_{a+b}(t))_{0 \leq a, b < m}$ for the full original reflected trace matrix. For real t the local test matrix is

$$C_\rho(t)_{al} = Z_{\rho, t}(F_a^\# F_l), \quad C_\rho(t) = R(t)^* K(t) R(t). \quad (\text{HR10})$$

This follows from HR2 and HR7 even at negative time: reflection permutes the nonreal roots, and remains the original coefficient conjugation. There is no replacement by a Euclidean norm there. For nonzero t , evaluation of the columns gives the Cauchy matrix $(1/(\rho + i\theta_\nu/2 - z_l))_{\nu l}$. Its determinant is the following exact formula:

$$\det R(t) = \frac{(-i/2)^M \prod_{a < b} (z_b - z_a)}{\prod_{\nu, l} (\rho + i\theta_\nu/2 - z_l)}. \quad (\text{HR11})$$

For completeness, multiplying the Cauchy determinant by its common denominator produces an alternating polynomial separately in the row points and in the poles. Its degrees force it to be a constant times the two Vandermonde products. Comparing one successive pole residue, starting with size one, gives the constant $(-1)^M$. The row Vandermonde is $(i/2)^M$ times the Vandermonde in θ_ν ; cancelling the latter with the evaluation determinant proves HR11. Both sides extend analytically to zero. Therefore the identity also follows there without a division by a vanishing Vandermonde.

The exact determinant and its first relative correction are

$$\det C_\rho(t) = \text{disc}_\xi W(t, \xi) \frac{4^{-M} |\prod_{a < b} (z_b - z_a)|^2}{|\prod_{\nu, l} (\rho + i\theta_\nu/2 - z_l)|^2}, \quad (\text{HR12})$$

$$\det C_\rho(t) = \frac{D_m 4^{-M} |\prod_{a < b} (z_b - z_a)|^2}{\prod_l |\rho - z_l|^{2m}} t^M (1 + \mathcal{E}_\rho t + O(t^2)), \quad D_m = 2^M \prod_{a=1}^m a^a, \quad (\text{HR13})$$

$$\mathcal{E}_\rho = m(m-1)b - 2mc \Re \left(i \sum_l \frac{1}{\rho - z_l} \right) - \frac{m(m-1)}{2} \Re \sum_l \frac{1}{(\rho - z_l)^2}. \quad (\text{HR14})$$

Indeed $\det K = \text{disc } W$ by the Vandermonde square. HC14 evaluates it as $D_m t^M (1 + m(m-1)bt + O(t^2))$. For $a_l = \rho - z_l$, HR1 gives

$$\sum_\nu \log(a_l + i\theta_\nu/2) = m \log a_l + t \left(\frac{imc}{a_l} + \frac{m(m-1)}{4a_l^2} \right) + O(t^2). \quad (\text{HR15})$$

Choose any local logarithm at each nonzero a_l ; its derivative and the real part are independent of that choice. Equations HR12–15 prove the claim. The full original unit enters through $b = 2d - c^2$ and c ; no model values are inserted.

HR5. Complete support and the arithmetic complement. For the complete support lattice L , lift the module isomorphism HR7 by $(x, 1_L) \mapsto (R(t)x, 1_L)$ and $(0, \lambda) \mapsto (0, \lambda)$ at every lower support label. Its inverse is the identical construction using $R(t)^{-1}$. They preserve addition, scalar action, and all zero supports directly. The reflected pairing retains the output support as well; a zero scalar value remains supported zero at the specified label. This explicitly relates the fixed Cauchy tests to the original full supported module, including its endpoint nilpotents.

To identify the exact place of the local matrix inside the whole arithmetic expression, use the original $g_t(s) = 16H_t(-2i(s-1/2))$ and $L_t = g'_t/g_t$. At zero,

$$L_0(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \psi(s/2) - \sum_{n \geq 2} \Lambda(n) n^{-s}, \quad \Re s > 1. \quad (\text{HR16})$$

This follows by the logarithmic derivative of $g_0 = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$; all series are absolutely convergent on the displayed domain. The complete Cauchy-test arithmetic matrix from SZW is

$$C_{\text{full}}(0)_{al} = \frac{L_0(z_l) + \overline{L_0(z_a)}}{z_l + \bar{z}_a - 1}, \quad C_{\text{ext}}(0) = C_{\text{full}}(0) - C_\rho(0), \quad C_\rho(0)_{al} = \frac{m}{\rho - z_a(\rho - z_l)}. \quad (\text{HR17})$$

For clarity, the first formula follows by subtracting the two genus-one logarithmic derivatives at z_l and $1 - \bar{z}_a$. Their constants and the regularizing $1/\rho'$ terms cancel, and each zero ρ' contributes $-1/((\rho' - (1 - \bar{z}_a))(\rho' - z_l))$, with its multiplicity. This is exactly $F_a^\#(\rho')F_l(\rho')$. The series converges absolutely since its terms are $O((1 + |\rho'|)^{-2})$ and the zero count is $O(R \log(2 + R))$, as proved in SZW25. Thus the subtraction does not discard any zero or any regularizing term. Reflection of the original completed zeta function gives $L_0(1 - \bar{z}_a) = -\overline{L_0(z_a)}$, which proves HR17. Every endpoint, Gamma term and prime power in HR16 is retained in C_{ext} by this exact subtraction. The local time derivative HR3 is likewise an actual part of the full derivative: the heat equation gives

$$J_0(s) = \partial_t L_t(s)|_0 = \frac{1}{4} \partial_s (L'_0(s) + L_0(s)^2), \quad \dot{C}_{\text{ext}}(0)_{al} = \frac{J_0(z_l) + \overline{J_0(z_a)}}{z_l + \bar{z}_a - 1} - \dot{C}_\rho(0)_{al}. \quad (\text{HR18})$$

The coordinate gives $\partial_t g = g''/4$, proving HR18 by differentiating the quotient at fixed poles, where g_0 is nonzero. The whole moving-zero interpretation follows from the complete companion proof `ACTUAL_HEAT_ZERO_DISTRIBUTION_VARIATION.md`, AG1–11, included unchanged in this package. Its actual heat integral gives uniform growth $|g_t(s)| \leq \exp(C_T(R+1)\log(R+2))$ on $|s| \leq R, |t| \leq T$. Jensen's formula and $g_t(0) \neq 0$ near zero give $N_t(R) = O(R \log(R+2))$ and the uniform $|\rho'|^{-2}$ tail. The genus-one logarithmic derivative then gives, for every rational $A = O(s^{-2})$ with its poles away from the zeros, the finite identity $Z_t(A) = -\sum_p \text{Res}_p(AL_t)$. Its right side is holomorphic in t , so the derivative in HR18 is the derivative of the actual whole trace. All growth estimates, pole exclusions, residue cancellations and the extension from original compact Mellin tests are proved in AG1–16; no derivative of an unrestricted infinite sum is assumed here. The finite local derivative is HR4 at $j = 0$. These formulas specify the full compensating matrix, rather than assigning its sign from the positive local flag coefficients. The lower support entries in SZW33–34 are fixed endpoint values; the present maps preserve them and their zero time derivative.

The classical factorization input in AG7 is Jacques Hadamard's finite-order factorization theorem; historical source: *Étude sur les propriétés des fonctions entières et en particulier d'une fonction considérée par Riemann*, Journal de Mathématiques Pures et Appliquées (4) 9 (1893), 171–215. The zero-count input is Johan Ludwig William Valdemar Jensen, *Sur un nouvel et important théorème de la théorie des fonctions*, Acta Mathematica 22 (1899), 359–364. The historical metadata are attribution, not a claim to have read original-author TeX for those nineteenth-century sources. The actual programme growth hypotheses and residue applications are proved in AG1–16.

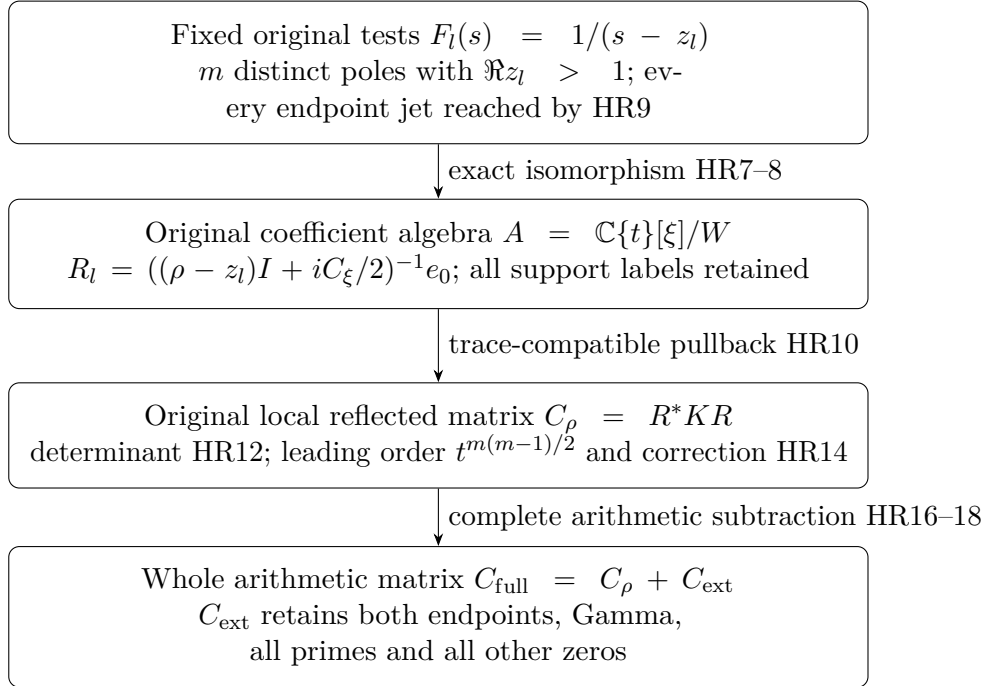


Figure 1: Fixed tests reach every original collision jet without changing the heat clock or the coordinate $s = \rho + i\xi/2$. The exact change of basis retains the full root polynomial and every support label. The local determinant is evaluated; the complete arithmetic matrix keeps its explicitly specified complement. Proofs HR1–18; human heat and explicit-formula sources are cited at the start.